SUBQUERY:

```
lab_15=# SELECT first_name, last_name, salary
lab_15-# FROM employees
lab_15-# WHERE salary > (
lab_15(# SELECT AVG(salary) -- This scalar subquery returns a single value
lab_15(# FROM employees
lab_15(# );
 first_name | last_name | salary
------------+-----------+---------
 Altynai    | Sultanova | 9000.00
 Bakyt      | Ismailov  | 7500.00
 Chyngyz    | Tokoev    | 7000.00
 Dinara     | Asanova   | 6800.00
 Erlan      | Kadyrov   | 5200.00
 Saltanat   | Bekova    | 5100.00
(6 rows)
```

Common Table Expressions (CTEs):

```
lab_15=# WITH regional_sales AS (
lab_15(# SELECT region_id, SUM(amount) AS total_sales
lab_15(# FROM orders
lab_15(# GROUP BY region_id
lab_15(# )
lab_15-# SELECT region_id, total_sales
lab_15-# FROM regional_sales
lab_15-# WHERE total_sales > 1000000;
 region_id | total_sales
-----------+-------------
         3 |  1530000.00
         4 |  1200000.00
         1 |  1100000.00
(3 rows)
```

RECURSIVE QUERIES:

```
lab_15=# WITH RECURSIVE org_chart AS (
lab_15(# SELECT employee_id, first_name, last_name, manager_id
lab_15(# FROM employees
lab_15(# WHERE manager_id IS NULL
lab_15(# UNION ALL
lab_15(# SELECT e.employee_id, e.first_name, e.last_name, e.manager_id
lab_15(# FROM employees e
lab_15(# INNER JOIN org_chart oc ON e.manager_id = oc.employee_id
lab_15(# )
lab_15-# SELECT * FROM org_chart;
 employee_id | first_name |   last_name    | manager_id
-------------+------------+----------------+------------
           1 | Altynai    | Sultanova      |
           2 | Bakyt      | Ismailov       |          1
           3 | Chyngyz    | Tokoev         |          1
           4 | Dinara     | Asanova        |          1
          16 | Perizat    | Alieva         |          1
           5 | Erlan      | Kadyrov        |          2
           6 | Fatima     | Zhunusova      |          2
           9 | Irina      | Petrova        |          3
          10 | Jamilya    | Orozova        |          3
          13 | Marat      | Sydykov        |          3
          17 | Rustam     | Nurlanov       |         16
          18 | Saltanat   | Bekova         |          4
           7 | Gulnara    | Abdyrakhmanova |          5
           8 | Hasan      | Toktogulov     |          5
          11 | Kanat      | Mamytov        |          9
          12 | Lola       | Rahimova       |          9
          14 | Nazira     | Kenzhebaeva    |         13
          15 | Omurbek    | Dzhumabaev     |         13
          19 | Talant     | Usenov         |         18
          20 | Ulan       | Zhakypov       |         18
(20 rows)
```

SET OPERATIONS (UNION):

```
lab_15=# SELECT product_name FROM warehouse_1
lab_15-# UNION
lab_15-# SELECT product_name FROM warehouse_2;
 product_name
--------------
 Printer
 Mouse
 Tablet
 Laptop
 Headphones
 Webcam
 Keyboard
 Monitor
 Router
 Smartphone
(10 rows)
```

SET OPERATIONS (INTERSECT):

```
lab_15=# SELECT product_id FROM warehouse_1 WHERE quantity = 0
lab_15-# INTERSECT
lab_15-# SELECT product_id FROM warehouse_2 WHERE quantity = 0;
 product_id
------------
          2
          4
(2 rows)
```

WINDOW FUNCTIONS:

```
lab_15=# SELECT
lab_15-# first_name,
lab_15-# last_name,
lab_15-# department,
lab_15-# salary,
lab_15-# RANK() OVER (PARTITION BY department ORDER BY salary DESC) AS dept_salary_rank
lab_15-# FROM employees;
 first_name |   last_name    | department  | salary  | dept_salary_rank
------------+----------------+-------------+---------+------------------
 Bakyt      | Ismailov       | Engineering | 7500.00 |                1
 Erlan      | Kadyrov        | Engineering | 5200.00 |                2
 Fatima     | Zhunusova      | Engineering | 4800.00 |                3
 Gulnara    | Abdyrakhmanova | Engineering | 4500.00 |                4
 Hasan      | Toktogulov     | Engineering | 4200.00 |                5
 Dinara     | Asanova        | Finance     | 6800.00 |                1
 Saltanat   | Bekova         | Finance     | 5100.00 |                2
 Talant     | Usenov         | Finance     | 4000.00 |                3
 Ulan       | Zhakypov       | Finance     | 3600.00 |                4
 Perizat    | Alieva         | HR          | 4600.00 |                1
 Rustam     | Nurlanov       | HR          | 3700.00 |                2
 Altynai    | Sultanova      | Management  | 9000.00 |                1
 Marat      | Sydykov        | Marketing   | 5000.00 |                1
 Nazira     | Kenzhebaeva    | Marketing   | 4400.00 |                2
 Omurbek    | Dzhumabaev     | Marketing   | 3800.00 |                3
 Chyngyz    | Tokoev         | Sales       | 7000.00 |                1
 Irina      | Petrova        | Sales       | 4900.00 |                2
 Jamilya    | Orozova        | Sales       | 4300.00 |                3
 Lola       | Rahimova       | Sales       | 4300.00 |                3
 Kanat      | Mamytov        | Sales       | 3900.00 |                5
(20 rows)
```

Pivoting with Conditional Aggregation:

```
lab_15=# SELECT
lab_15-# product_name,
lab_15-# SUM(CASE WHEN year = 2022 THEN sales_amount ELSE 0 END) AS year_2022,
lab_15-# SUM(CASE WHEN year = 2023 THEN sales_amount ELSE 0 END) AS year_2023,
lab_15-# SUM(CASE WHEN year = 2024 THEN sales_amount ELSE 0 END) AS year_2024
lab_15-# FROM sales
lab_15-# GROUP BY product_name;
 product_name | year_2022 | year_2023 | year_2024
--------------+-----------+-----------+-----------
 Smartphone   | 200000.00 | 220000.00 | 260000.00
 Tablet       |  60000.00 |  75000.00 |  90000.00
 Keyboard     |  18000.00 |  21000.00 |  25000.00
 Monitor      |  45000.00 |  52000.00 |  48000.00
 Laptop       | 120000.00 | 135000.00 | 150000.00
 Headphones   |  33000.00 |  41000.00 |  38000.00
(6 rows)
```

Unpivoting with UNION ALL:

```
lab_15=# SELECT product_name, '2022' AS year, year_2022 AS sales_amount FROM pivoted_sales
lab_15-# UNION ALL
lab_15-# SELECT product_name, '2023' AS year, year_2023 AS sales_amount FROM pivoted_sales
lab_15-# UNION ALL
lab_15-# SELECT product_name, '2024' AS year, year_2024 AS sales_amount FROM pivoted_sales
lab_15-# ORDER BY product_name, year;
 product_name | year | sales_amount
--------------+------+--------------
 Headphones   | 2022 |     33000.00
 Headphones   | 2023 |     41000.00
 Headphones   | 2024 |     38000.00
 Keyboard     | 2022 |     18000.00
 Keyboard     | 2023 |     21000.00
 Keyboard     | 2024 |     25000.00
 Laptop       | 2022 |    120000.00
 Laptop       | 2023 |    135000.00
 Laptop       | 2024 |    150000.00
 Monitor      | 2022 |     45000.00
 Monitor      | 2023 |     52000.00
 Monitor      | 2024 |     48000.00
 Smartphone   | 2022 |    200000.00
 Smartphone   | 2023 |    220000.00
 Smartphone   | 2024 |    260000.00
 Tablet       | 2022 |     60000.00
 Tablet       | 2023 |     75000.00
 Tablet       | 2024 |     90000.00
(18 rows)
```

FILTER CLAUSE:

```
lab_15=# SELECT
lab_15-# employee_id,
lab_15-# SUM(salary) OVER (PARTITION BY department) AS total_dept_salary,
lab_15-# AVG(salary) FILTER (WHERE tenure > 5) OVER () AS avg_salary_senior
lab_15-# FROM employees;
 employee_id | total_dept_salary |   avg_salary_senior
-------------+-------------------+-----------------------
           6 |          26200.00 | 6122.2222222222222222
           7 |          26200.00 | 6122.2222222222222222
           8 |          26200.00 | 6122.2222222222222222
           2 |          26200.00 | 6122.2222222222222222
           5 |          26200.00 | 6122.2222222222222222
          19 |          19500.00 | 6122.2222222222222222
          20 |          19500.00 | 6122.2222222222222222
           4 |          19500.00 | 6122.2222222222222222
          18 |          19500.00 | 6122.2222222222222222
          17 |           8300.00 | 6122.2222222222222222
          16 |           8300.00 | 6122.2222222222222222
           1 |           9000.00 | 6122.2222222222222222
          13 |          13200.00 | 6122.2222222222222222
          14 |          13200.00 | 6122.2222222222222222
          15 |          13200.00 | 6122.2222222222222222
          12 |          24400.00 | 6122.2222222222222222
          11 |          24400.00 | 6122.2222222222222222
          10 |          24400.00 | 6122.2222222222222222
           9 |          24400.00 | 6122.2222222222222222
           3 |          24400.00 | 6122.2222222222222222
(20 rows)
```

Custom Sorting with CASE:

```
lab_15=# SELECT product_name, status
lab_15-# FROM orders
lab_15-# ORDER BY
lab_15-# CASE status
lab_15-# WHEN 'High Priority' THEN 1
lab_15-# WHEN 'Medium Priority' THEN 2
lab_15-# WHEN 'Low Priority' THEN 3
lab_15-# ELSE 4
lab_15-# END;
 product_name |     status
--------------+-----------------
 Laptop       | High Priority
 Tablet       | High Priority
 Smartphone   | High Priority
 Keyboard     | High Priority
 Laptop       | High Priority
 Smartphone   | Medium Priority
 Printer      | Medium Priority
 Laptop       | Medium Priority
 Headphones   | Medium Priority
 Tablet       | Low Priority
 Monitor      | Low Priority
 Monitor      | Low Priority
 Keyboard     | Low Priority
 Headphones   | Pending
 Printer      | Pending
(15 rows)
```

OPTIMIZATION: EXPLAIN

```
lab_15=# EXPLAIN
lab_15-# SELECT first_name, last_name, salary
lab_15-# FROM employees
lab_15-# WHERE salary > (
lab_15(# SELECT AVG(salary)
lab_15(# FROM employees
lab_15(# );
                                  QUERY PLAN
-------------------------------------------------------------------------------
 Seq Scan on employees  (cost=12.39..24.76 rows=63 width=252)
   Filter: (salary > (InitPlan 1).col1)
   InitPlan 1
     ->  Aggregate  (cost=12.38..12.39 rows=1 width=32)
           ->  Seq Scan on employees employees_1  (cost=0.00..11.90 rows=190 width=16)
(5 rows)
```

OPTIMIZATION: EXPLAIN ANALYZE

```
lab_15=# EXPLAIN ANALYZE
lab_15-# SELECT first_name, last_name, salary
lab_15-# FROM employees
lab_15-# WHERE salary > (
lab_15(# SELECT AVG(salary)
lab_15(# FROM employees
lab_15(# );
                                                    QUERY PLAN
-------------------------------------------------------------------------------------------------------------------------------------------
 Seq Scan on employees  (cost=12.39..24.76 rows=63 width=252) (actual time=0.058..0.068 rows=6.00 loops=1)
   Filter: (salary > (InitPlan 1).col1)
   Rows Removed by Filter: 14
   Buffers: shared hit=2
   InitPlan 1
     ->  Aggregate  (cost=12.38..12.39 rows=1 width=32) (actual time=0.032..0.032 rows=1.00 loops=1)
           Buffers: shared hit=1
           ->  Seq Scan on employees employees_1  (cost=0.00..11.90 rows=190 width=16) (actual time=0.010..0.015 rows=20.00 loops=1)
                 Buffers: shared hit=1
 Planning Time: 0.179 ms
 Execution Time: 0.102 ms
(11 rows)
```

OPTIMIZATION: LIMIT

```
lab_15=# SELECT first_name, last_name, salary
lab_15-# FROM employees
lab_15-# ORDER BY salary DESC
lab_15-# LIMIT 10;
 first_name | last_name | salary
------------+-----------+---------
 Altynai    | Sultanova | 9000.00
 Bakyt      | Ismailov  | 7500.00
 Chyngyz    | Tokoev    | 7000.00
 Dinara     | Asanova   | 6800.00
 Erlan      | Kadyrov   | 5200.00
 Saltanat   | Bekova    | 5100.00
 Marat      | Sydykov   | 5000.00
 Irina      | Petrova   | 4900.00
 Fatima     | Zhunusova | 4800.00
 Perizat    | Alieva    | 4600.00
(10 rows)
```